

RFID/NFC Based Attendance System with Cloud Integration: A Scalable and Cost-Effective Approach for Educational Institutions.

INTRODUCTION

- 1. Problem:**

Many government schools still use paper registers or expensive fingerprint scanners. These methods are slow, prone to mistakes, and often break down, leading to inaccurate records.
- 2. Why This Solution:**

There is a clear need for a system that is both affordable and reliable, reduces teacher workload, and delivers accurate attendance instantly.
- 3. Product Solution:**

The proposed cloud-connected RFID/NFC attendance system that employs a PN532 reader and a Node-MCU ESP8266. Students tap inexpensive NFC tags, and attendance is captured automatically.
- 4. Key Result:**

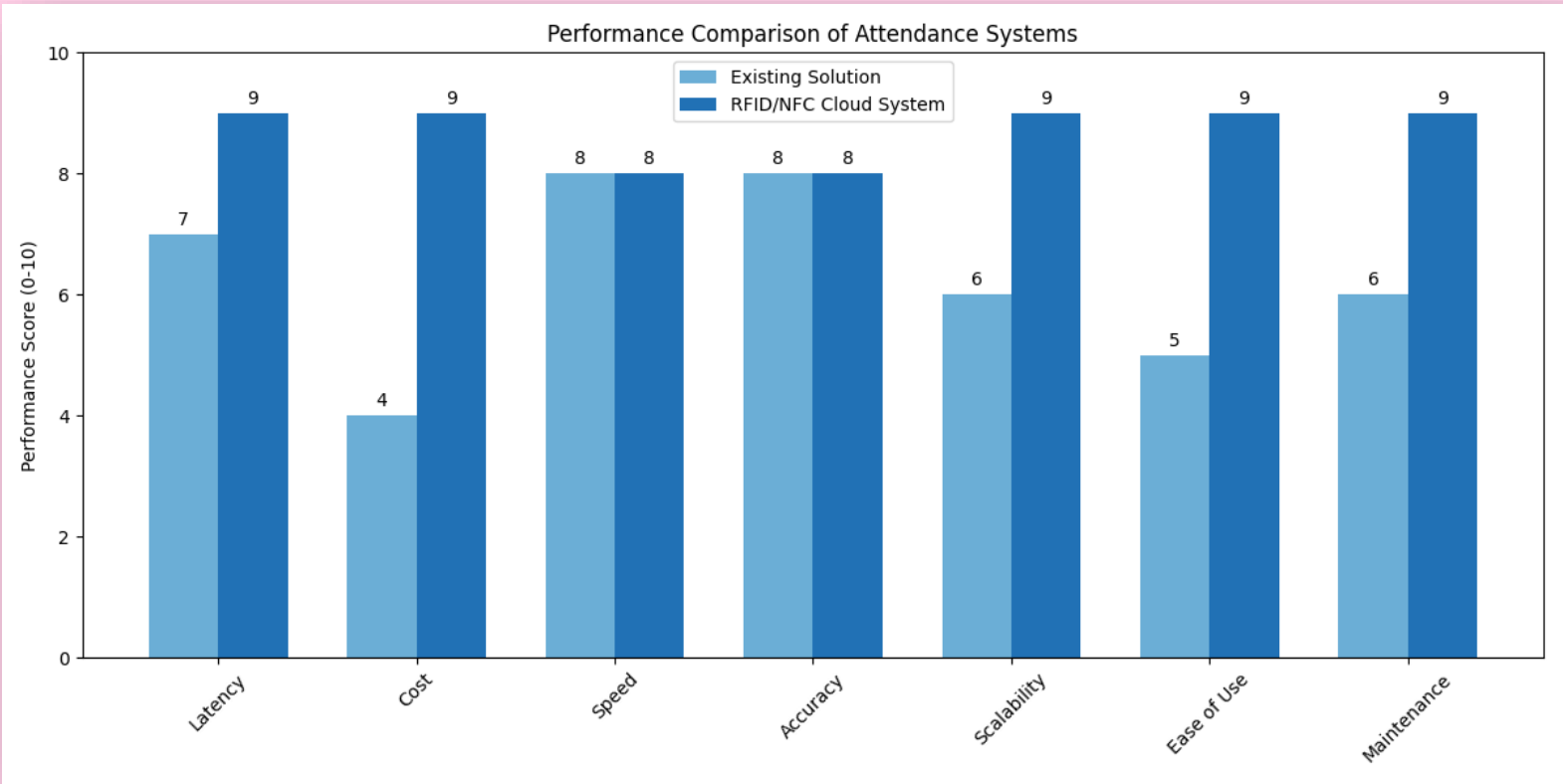
In tests, our system records each tap in under 2 seconds, pushes data to the cloud with almost zero delay, and cuts overall costs by around 30 % compared to fingerprint devices.

MATERIALS AND METHODS

- 1. Hardware Components:**
 - PN532 RFID/NFC Reader: Reads student tags quickly and reliably.
 - Node-MCU ESP8266: Processes tag IDs and handles Wi-Fi communication.
 - NFC Tags: Low-cost cards assigned to each student.
- 2. Software & Cloud Setup:**
 - Arduino IDE: Code for NFC reading and Wi-Fi routines
 - Google Sheets API: Secure endpoint for real-time data storage.
- 3. Workflow:**
 - Student taps NFC tag on the PN532 reader.
 - Node-MCU captures tag ID and timestamp.
 - Data is sent over Wi-Fi to a shared Google Sheet.
 - Administrators access and manage attendance online instantly.

RESULTS

Parameter	Manual Register	Biometric System	RFID/NFC Cloud System
Cost	Low hardware cost but high labour expenses	Very high setup and maintenance costs	Very low hardware + free cloud storage
Speed (per student)	10–20 s (manual entry)	2–5 s (scanning)	1–2 s (tap & cloud update)
Accuracy	Error-prone (missing entries)	Generally accurate but sensor glitches can occur	Automated read, near-zero error rate
Latency	Minutes to hours (data entry delays)	5–10 s (processing overhead)	1-2 s (real-time cloud sync)
Scalability	Not suitable for storage efficient	Increased hardware cost per location	Easily add readers and tags, no extra infrastructure



DISCUSSION AND CONCLUSION

- Discussion:**
 - Efficiency: Teachers save valuable class time—no manual roll-calls or waiting for scanners.
 - Reliability: NFC tags and cloud integration eliminate human error and data loss.
 - Affordability: Low-cost tags and open-source hardware cut initial and ongoing expenses.
 - Ease of Management: A single Google Sheet replaces bulky paperwork and complex databases.
- Conclusion:**

The proposed cloud-connected RFID/NFC attendance system offers a practical, scalable, and budget-friendly solution for schools. It streamlines attendance tracking, delivers real-time data to administrators, and significantly reduces both administrative workload and costs. This makes it an ideal choice for government and resource-constrained educational environments.

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